features learnt belong to class 0 only.

- *Class 1:* In these feature maps, the features learnt belong to class 1 only.
- *Mixed:* In these feature maps, the features learnt belong to both class 0 and class 1.
- *Inactive:* These feature maps do not learn any features of class 0 or class 1.

The type of activation feature maps in each layer can be easily visualized using the Deepvis tool specified above. Many a times, complex standard models are used to build a simple binary as above, or a model for up to ten class classifiers using transfer learning. During those scenarios, as seen above, many filters may not be required since they will not be learning any features at all, and fewer filters will be sufficient to learn features of all the classes. In these cases, if the requirement is to retain the original model architecture for some reason, then the inactive filters can be removed to reduce the computations and the size of the model.

The types of filters will also help to debug when the test image is mis-classified. The types of filters at each layer can be studied for both the classes during training and during evaluation. The filters activated can be compared during training and evaluation. When making a correct prediction of the test image, the count of the type of filters for each class will approximately match the average count of the type of filters for each class during training. This means that if a test image is correctly classified as a passenger flight class, then its count of the type of filters (class 0, class 1, mixed and inactive) must be approximately equal to the average count of the type of filters of the passenger flight determined during training, and the same will be true for an image correctly classified as the fighter flight class.

These findings can help to debug mis-classified images. For example, if the test image is mis-classified as the fighter class, then the count of the type of filters will not match the count of the type of filters for the fighter class during training, which is quite obvious. The potential reason for mis-classification could be any of the following:

- The model has evaluated the features of both the fighter and passenger flight classes, and found that fighter features dominating the passenger features during classification and hence it is mis-classified.
- The model has evaluated and found more of fighter features matching during classification and hence it is mis-classified.
- The model has evaluated and found comparable fighter and passenger features matching during classification and due to minor differences in features learnt, it has been mis-classified

To understand the above reasoning, the activation feature maps of each convolution layer need to be analysed, and the average count of the type of filters at each layer may have to be considered to come to a conclusion. There are approaches to handle this, which will be discussed in the next article.

This article provides an overview of how the visual cortex perceives objects. It explains the parallels between CNNs and different building blocks of the CNN architecture. It also briefly covers each building block and how the convolution layer processes the image features from the input. It then highlights the importance of activation during image classification. Knowing the details about the activations of each filter in each layer allows model developers to understand the importance of each filter and update their CNN architectures or build new CNNs with reduced parameters and computations. The article also gives details on the types of activation feature maps, and how they can be used for debugging CNN models and to detect potential mis-classification. Further exploration and research on activation feature maps for different types of models other than CNN will provide many insights, especially for object detection and recognition models. The same can be extended to natural language processing and voice recognition models, the activation feature maps for which would be a bit more complex to analyse. END

References

[1] *https://towardsdatascience.com/applied-deep-learning-part-4-convolutional-neural-networks-584bc134c1e2*
[2] *http://yosinski.com/deepvis*
[3] *https://towardsdatascience.com/wtf-is-image-classification-8e78a8235acb*
[4] *https://www.tensorflow.org/*

By: B.N. Chandrashekar, Dr Manjunath Ramachandra and Shashidhar Soppin

B.N. Chandrashekar is a principal consultant and researcher with hands-on programming experience. He has around two decades of industry experience. With a Masters from IISc, he specialises in embedded, retail, e-commerce and AI driven technology.

Dr Manjunath Ramachandra has over two decades of work experience in the overlapping verticals of artificial intelligence, computer vision, healthcare and wireless/ mobile technologies. He figures in the list of the '2000 outstanding intellectuals of the 21st century', brought out by the International Biographical Center, UK.

Shashidhar Soppin, DMTS (distinguished member of the technical staff) senior member of Wipro, has 19+ years of experience in the IT industry. He specialises in virtualisation, Docker, the cloud, AI, ML, deep learning and OpenStack.